MIL-R-82483A(OS)
10 March 1976
SUPERSEDING
MIL-R-82483(OS)
16 November 1967

# MILITARY SPECIFICATION

# RESIN, POLYESTER, SOLUTIONS

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all the departments and agencies of the Department of Defense.*

## 1. SCOPE

1.1 Scope. This specification covers polyester resin solutions for use in explosives.

1.2 Classification. The polyester resin solution shall be of the following types as specified (see 6.2):

Type I - Styrene solution (46 percent nonvolatile)
Type II - Ethyl acetone solution (33 percent nonvolatile)

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATIONS

Federal

| | |
|---|---|
| TT-E-751 | Ethyl Acetate, Technical |
| TT-P-143 | Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of |

FSC 6810

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Military

| | |
|---|---|
| MIL-S-14195 | Styrene Monomer |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-141 | Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing |

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

American Society for Testing and Materials (ASTM)

| | |
|---|---|
| D 638 | Test for Tensile Properties of Plastics |
| D 695 | Test for Compressive Properties of Rigid Plastic |
| D 790 | Test for Flexural Properties of Plastics |
| D 890 | Test for Water in Liquid Naval Stores |



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(Copies of ASTM publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

## 3. REQUIREMENTS

3.1 Resin characteristics. The resin component in both types I and II solutions shall conform to the requirements listed in table I when determined at room temperature on 1/8-inch nonreinforced castings made from the basic resin solids (46 percent) and styrene (54 percent).

TABLE I. Resin requirements.

| Test | Requirement | | Method paragraph |
|---|---|---|---|
| | Minimum | Maximum | |
| Flexural strength, psi | 9,500 | 13,500 | 4.4.1 |
| Flexural modulus, psi × $10^6$ | 0.25 | 0.35 | 4.4.1 |
| Tensile strength, psi | 5,000 | 7,000 | 4.4.2 |
| Elongation, % | 4.0 | 5.0 | 4.4.2 |
| Compressive strength, psi | 9,000 | 11,000 | 4.4.3 |

3.2 Type I solution. The type I solution shall consist of 46 percent by weight of the resin solids dissolved in 54 percent by weight of styrene monomer conforming to MIL-S-14195 (with an inhibitor level of up to 50 parts per million permissible). The solution shall conform to the requirements listed in table II.

3.3 Type II solution. The type II solution shall consist of 33 percent by weight of the resin solids and 67 percent by weight of ethyl acetate conforming to TT-E-751. The solution shall conform to the requirements listed in table III.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government



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reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

TABLE II. Type I solution requirements.

| Test | Requirement | | Method paragraph |
|---|---|---|---|
| | Minimum | Maximum | |
| Nonvolatile content, percent by weight | 44.0 | 48.0 | 4.4.4 |
| Viscosity, poises | - | 0.45 | 4.4.5 |
| Specific gravity, 25° C | 1.01 | 1.05 | 4.4.6 |
| Acid number | - | 15 | 4.4.7 |
| Color, APHA | - | 100 | 4.4.8 |
| Phthalic anhydride, percent by weight of resin solids | 23 | 27 | 4.4.9 |
| Water content, Karl Fischer, percent | - | 0.20 | 4.4.10 |
| Gel time, 50° C, minutes | 10 | 20 | 4.4.11 |

TABLE III. Type II solution requirements.

| Test | Requirement | | Method paragraph |
|---|---|---|---|
| | Minimum | Maximum | |
| Nonvolatile content, percent by weight | 32 | 34 | 4.4.4 |
| Viscosity, poises | - | 0.3 | 4.4.5 |
| Acid number | - | 15 | 4.4.7 |
| Color, APHA | - | 70 | 4.4.8 |
| Phthalic anhydride, percent by weight of resin solids | 23 | 27 | 4.4.9 |
| Water content, Karl Fischer, percent | - | 0.20 | 4.4.10 |



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4.2 Quality conformance inspection. The quality conformance inspection shall include an examination for acceptability of facilities and quality control methods used by the manufacturer, examining and testing the acceptance samples (4.3.1) for conformance to all the requirements of this specification, and shall include an examination of the samples of filled containers for conformance to packaging, packing, and marking requirements.

4.3 Sampling.

4.3.1 Quality conformance sample. The quality conformance sample shall consist of a sample for tests (4.3.1.2) and samples for examination of filled containers (4.3.1.3). Samples shall be labeled completely with information identifying the purpose of the sample, name of product, specification number, lot and batch number, date of sampling, and contract number.

4.3.1.1 Lot and batch. All resin solution manufactured as one batch shall be considered a lot and numbered as such for purposes of inspection. A batch is defined as the end product of all the raw materials mixed or blended in a single operation.

4.3.1.2 Samples for tests. The samples for tests specified in 4.4 shall consist of 1 gallon of resin solution taken at random from each lot.

4.3.1.3 Sample for examination of filled containers. A random sample of filled containers and a sample of shipping containers fully prepared for delivery shall be selected from each lot of resin solution in accordance with MIL-STD-105 at inspection level I and acceptable quality level (AQL) = 2.5 percent defective and examined for fill, closure, and marking.

4.4 Test methods.

4.4.1 Flexural strength and flexural modulus. Flexural strength and flexural modulus shall be determined in accordance with ASTM D 790, procedure B.

4.4.2 Tensile strength and elongation. Tensile strength and elongation shall be determined in accordance with ASTM D 368. Speed of testing shall be 0.2 inch per minute.



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4.4.3 Compressive strength. Compressive strength shall be determined in accordance with ASTM D 695.

4.4.4 Nonvolatile content. For type I resin solution, 1 gram is weighed accurately into a tared aluminum weighing dish (flat bottom) and then 10 grams of ethyl acetate added to dilute the resin. At least two samples should be run at the same time. The dishes are then heated in an air circulating oven for 2 hours at 105° C. This removes all the ethyl acetate and styrene monomer in the resin, leaving the alkyd or solids portion of the resin. The aluminum dishes are reweighed and the volatile content determined by the difference from the original. The original sample weight minus the volatile weight content is the nonvolatile or solids content. For type II resin solution, the determination shall be conducted as above except that no ethyl acetate shall be added.

4.4.5 Viscosity. The viscosity of the resin solution shall be determined in accordance with method 4271 of FED-STD-141 or any other suitable method.

4.4.6 Specific gravity. The specific gravity shall be determined in accordance with method 4184 of FED-STD-141 or any other suitable method.

4.4.7 Acid number. The acid number shall be determined in accordance with 5073 of FED-STD-141. Calculate acid number on the basis of the entire sample of type I or type II resin. Modify procedure in method 5073 when using type II solution as follows: After weighing a sample of the solution, remove the ethyl acetate by heating the solution at 105° C for 2 hours before proceeding with the remainder of the determination.

4.4.8 Color. The color of the resin solution shall be determined using the following method.

4.4.8.1 Apparatus. Colorimeter equivalent to Bausch and Lomb Duboscq Calorimeter. The instrument shall have a light source equivalent to "North Sky" illumination.

4.4.8.2 Standard color solution. Dissolve 1.246 grams of potassium chloroplatinate ($K_2PtCl6$) and 1.0 gram of cobaltous chloride ($CoCl_2.6H_2O$) in 100 milliliter (ml) of water and 100 ml of concentrated hydrochloric acid. Dilute the mixture to 1 liter in a volumetric flask. For each



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number of degree of color, add 0.2 ml of the standard color solution to a Nessler tube and make up to 100-ml mark with distilled water. Thus for a No. 10 color add 2 ml of standard color solution and dilute to the mark with water. Keep all comparison tubes covered to prevent introduction of dirt.

4.4.8.3 Procedure for type I solution. Fill the standard cup with APHA Color No. 100 solution. Set at 15 on the instrument scale. Fill the other cup with type I solution. Turn its scale until the light balances with the standard. Record the scale reading. Solve for the color number by the equation:

$$C = A \times \frac{B}{D}$$

where:

C = Color No. of sample
A = Color No. of standard
B = Scale setting of standard (15)
D = Scale setting of sample.

4.4.8.4 Procedure for type II solution. Same as 4.4.8.3 except use a standard of APHA Color No. 70.

4.4.9 Phthalic anhydride. Phthalic anhydride content shall be determined in accordance with method 7022 of FED-STD-141.

4.4.10 Water content, Karl Fischer. Karl Fischer water content shall be detemmined in accordance with ASTM D 890.

4.4.11 Gel time. A 100-gram sample of resin is weighed into a 200-ml tall form beaker and the resin sample brought to 50° C. A 0.13 percent cobalt napthenate solution, 6 percent cobalt, is added to the resin and stirred in thoroughly. When the resin has reached a uniform pink or red color, 0.26 percent (0.26 ml) of methyl ethyl ketone peroxide catalyst is added and the solution stirred for at least 1 minute. The time of addition of peroxide is taken as the starting time for the gel determination. The resin is maintained at 50° C, with a stirring rod left in the beaker. The rod is periodically withdrawn from the resin to test for gel formation. The resin is considered gelled when it "snaps back" on



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removal of the stirring rod. The gel time is the time from catalyst addition to the gel of the resin.

4.5 Rejection criteria. If a test sample fails to meet any of the requirements of this specification, the lot represented by the sample shall be rejected.

5. PACKAGING

5.1 The resin solutions shall be packaged, packed, and marked for shipment in accordance with TT-P-143. Quantity per container shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The resin solutions covered by this specification are intended for use in explosives.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type of resin solution required
c. Size of container (see 5.1)
d. Quantity required (see 5.1).

6.3 Resin solution should be purchased by net weight.

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DOCUMENT IDENTIFIER AND TITLE

MIL-R-82483A(OS) Resin, Polyester, Solutions

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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